

EC466/966: Estimation and Inference in Econometrics

Answer key-Exercise # 2

1. Please check EC966_466hw2sol.py.
2. Suppose that you have the following regression model

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times K$ vector of regressors, β is the $K \times 1$ vector of the model parameters, and ϵ is the $n \times 1$ vector of the error term. Assume that Classical Linear Regression Model assumptions hold. We estimate the model by the OLS estimator, and obtain $\mathbf{b}$ that is the estimate for β .

- (a) Show that $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$ where $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

We use the definitions of $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$ and $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$; therefore we write that

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{b} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{P}\mathbf{y}.$$

- (b) Show that $\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}\epsilon$ where $\mathbf{M} = \mathbf{I}_n - \mathbf{P}$ and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.

First, we write that

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{P}\mathbf{y} = (\mathbf{I}_n - \mathbf{P})\mathbf{y} = \mathbf{M}\mathbf{y}.$$

Next, we use the identity for $\mathbf{y} = \mathbf{X}\beta + \epsilon$ and write that

$$\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mathbf{X}\beta + \epsilon) = \mathbf{M}\mathbf{X}\beta + \mathbf{M}\epsilon.$$

Note that $\mathbf{M}\mathbf{X} = (\mathbf{I}_n - \mathbf{P})\mathbf{X} = \mathbf{X} - \mathbf{P}\mathbf{X} = \mathbf{X} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{0}$. So, we conclude that

$$\mathbf{e} = \mathbf{M}\mathbf{X}\beta + \mathbf{M}\epsilon = \mathbf{M}\epsilon.$$

(c) Show that $SSR(\mathbf{b}) = \epsilon' \mathbf{M} \epsilon$.

We know that $SSR(\mathbf{b}) = \mathbf{e}' \mathbf{e}$. By using the identity in the above Question 2(b) such that $\mathbf{e} = \mathbf{M} \epsilon$.

$$SSR(\mathbf{b}) = (\mathbf{M} \epsilon)' \mathbf{M} \epsilon = \epsilon' \mathbf{M}' \mathbf{M} \epsilon.$$

Note that $\mathbf{M}$ is a symmetric matrix, since $\mathbf{M}' = (\mathbf{I}_n - \mathbf{P})' = \mathbf{I}'_n - \mathbf{P}' = \mathbf{I}'_n - (\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')' = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{M}$. Also, $\mathbf{M}$ is an idempotent matrix since

$$\mathbf{M}\mathbf{M} = (\mathbf{I}_n - \mathbf{P})(\mathbf{I}_n - \mathbf{P}) = \mathbf{I}_n - 2\mathbf{P} + \mathbf{P}\mathbf{P} = \mathbf{I}_n - 2\mathbf{P} + \mathbf{P} = \mathbf{I}_n - \mathbf{P},$$

where $\mathbf{P}\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{P}$. By using $\mathbf{M}$ being a symmetric and idempotent matrix, we can write that

$$SSR(\mathbf{b}) = \epsilon' \mathbf{M}' \mathbf{M} \epsilon = \epsilon' \mathbf{M} \mathbf{M} \epsilon = \epsilon' \mathbf{M} \epsilon.$$

(d) Show that $s^2 = \mathbf{e}' \mathbf{e} / (n - K)$ is an unbiased estimator for $\sigma^2 = E(\epsilon_i^2 | \mathbf{X})$.

To show that s^2 is an unbiased estimator for σ^2 , we need to prove that $E(s^2 | \mathbf{X}) = \sigma^2$.

$$\begin{aligned} E(s^2 | \mathbf{X}) &= (n - K)^{-1} E(\mathbf{e}' \mathbf{e} | \mathbf{X}) = (n - K)^{-1} E(\epsilon' \mathbf{M} \epsilon | \mathbf{X}) \\ &= (n - K)^{-1} E(tr(\epsilon' \mathbf{M} \epsilon) | \mathbf{X}) = (n - K)^{-1} E(tr(\mathbf{M} \epsilon \epsilon') | \mathbf{X}) \\ &= (n - K)^{-1} tr(E(\mathbf{M} \epsilon \epsilon' | \mathbf{X})) = (n - K)^{-1} tr(\mathbf{M} E(\epsilon \epsilon' | \mathbf{X})) \\ &= (n - K)^{-1} tr(\mathbf{M} \sigma^2 \mathbf{I}_n) = \sigma^2 (n - K)^{-1} tr(\mathbf{M}) \\ &= \sigma^2 (n - K)^{-1} tr(\mathbf{I}_n - \mathbf{P}) = \sigma^2 (n - K)^{-1} (n - tr(\mathbf{P})) \\ &= \sigma^2 (n - K)^{-1} (n - tr(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')) \\ &= \sigma^2 (n - K)^{-1} (n - tr((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X})) \\ &= \sigma^2 (n - K)^{-1} (n - tr(\mathbf{I}_K)) \\ &= \sigma^2. \end{aligned}$$